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This Detroit startup is turning to utilities to make home efficiency upgrades cheaper

Tim De Chant — 6:32 AM PDT · August 1, 2025

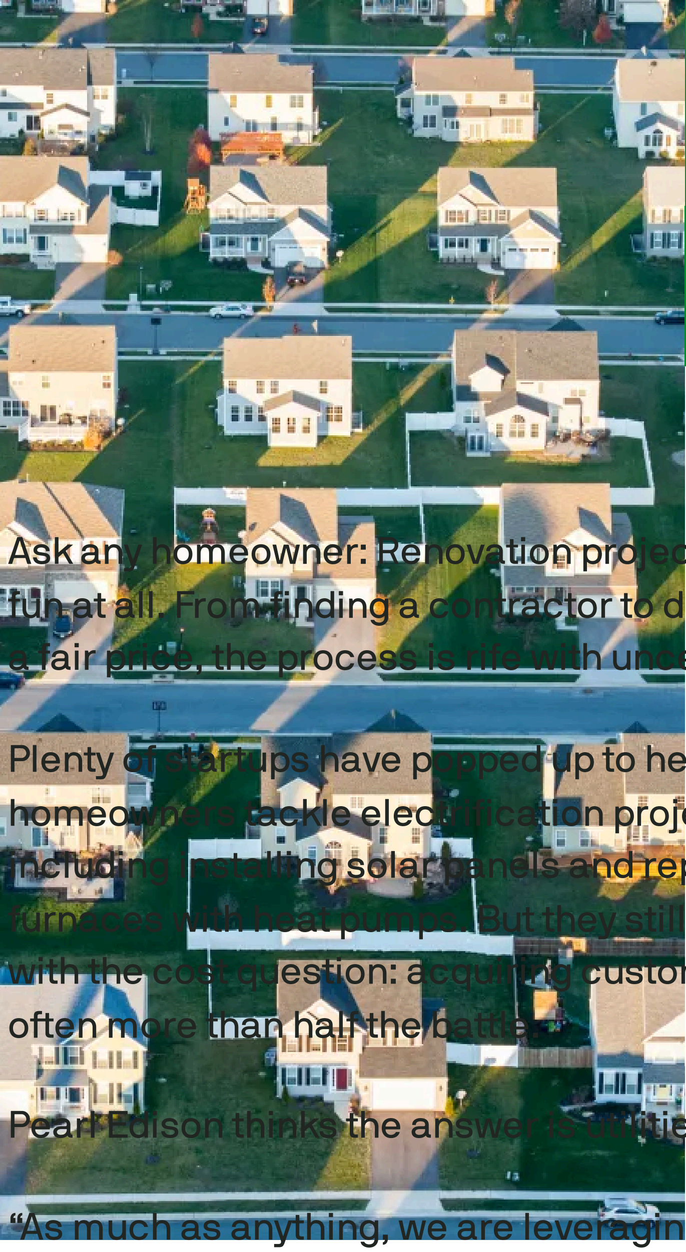

IMAGE CREDITS: EDWIN REMSBERG / GETTY IMAGES

Ask any homeowner: Renovation projects are no fun at all. From finding a contractor to determining a fair price, the process is rife with uncertainty.

Plenty of startups have popped up to help homeowners tackle electrification projects, including installing solar panels and replacing gas furnaces with heat pumps. But they still struggle with the cost question: acquiring customers is often more than half the battle.

Pearl Edison thinks the answer is utilities.

"As much as anything, we are leveraging their brand equity and trust," Evan Anderson, co-founder of Pearl Edison, told TechCrunch.

Investors have bought into the idea. The Detroit-based startup has raised a $3.3 million seed round from New System Ventures and Commonweal Ventures, the company exclusively told TechCrunch. Lightbank and Newlab participated.

Pearl Edison works with utilities to identify customers who are most likely to benefit from energy retrofit projects, including heat pumps and additional insulation. It builds a white-labeled website for the utility, and it helps the larger organization run campaigns to sign people up for energy efficiency upgrades.

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The company uses a range of data sources to design a job and generate a price, which it guarantees for the customer. After its software produces a first draft of the plans, it sends workers out to verify everything in the field before finalizing the design.

It then sends the jobs to vetted contractors, who complete the installation. Pearl Edison makes money on the installs itself, estimating that the majority of those jobs will cost less than it quotes. Homeowners should save money, too, Anderson said, because Pearl Edison can contract the job for less since the installers don't have to waste time acquiring customers.

Already, the company has programs set up with two utilities, DT Energy in Michigan and Duquesne Light in the Pittsburgh area, and one government, the city of Ann Arbor, Michigan. Anderson said the company will add two more utilities this year.

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"We've found them to be good partners," Anderson said.

Topics: Climate energy efficiency Fundraising heat pumps housing solar panels utilities

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Elon Musk is probably right about one thing

Tim De Chant — 12:15 PM PDT · July 31, 2023

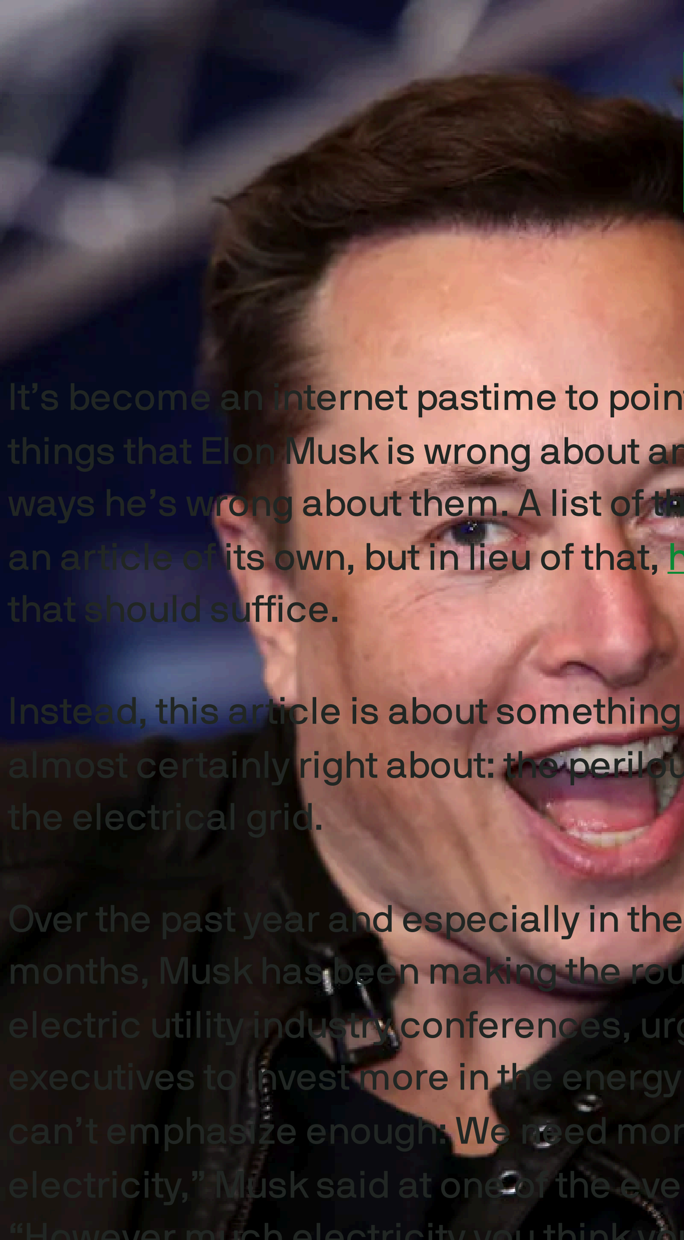

IMAGE CREDITS: LIESA JOHANNSSEN-KOPPITZ/BLOOMBERG / GETTY IMAGES

It's become an internet pastime to point out all the things that Elon Musk is wrong about and all the ways he's wrong about them. A list of them could be an article of its own, but in lieu of that, here's a link that should suffice.

Instead, this article is about something that Musk is almost certainly right about: the perilous state of the electrical grid.

Over the past year and especially in the past couple months, Musk has been making the rounds at electric utility industry conferences, urging executives to invest more in the energy transition. "I can't emphasize enough: We need more electricity," Musk said at one of the events. "However much electricity you think you need, more than that is needed."

Electric utilities have been anticipating a surge in demand as governments and consumers grow increasingly bullish on electrification. Homes, business and transportation all stand to benefit from more powerful and more efficient batteries, electric motors, heat pumps and more. In California, which is at the forefront of electrification, PG&E is predicting that electricity demand will rise 70% by 2043; Edison International is forecasting 60% growth by 2045.

Musk has said those predictions are far too modest. "So even if you assume that electricity demand is static, in order to transition to a sustainable energy future where everything is electric and sustainably electric, we need a tripling of electrical output," he said at another event.

Energy agencies have been notorious for underestimating the speed at which renewables would be adopted, in part because their forecasts have been conservative about cost decreases while also assuming that current regulatory policies won't get any more stringent.

The U.S. Energy Information Administration (EIA) and the International Energy Association (IEA) have both been guilty of that. For years, the IEA took the most recent data on wind and solar and assumed linear growth when the preceding years showed an exponential relationship. (The IEA has since added a net-zero emissions report in an apparent effort to hedge its bets.) Meanwhile, the EIA still assumes the U.S. will be burning coal well past 2050.

It's likely that utilities are guilty of similar thinking. With billions invested in legacy fossil fuel assets, they're probably assuming they'll be operating those far longer than Musk has in mind. Unless the world spends far more than it should on carbon capture and sequestration, coal burning power plants are going to have to disappear. They're obviously incompatible with the idea of reaching net-zero by 2050.

Musk is likely assuming that all combustion generators, including gas plants running on landfill gas and hydrogen, will be shut down. Wind, solar, batteries, and other carbon-free sources would take their place.

Musk is well known to approach problems from a "first principles" perspective. In this case, that probably means he's starting from the assumption that the world will have to hit net-zero emissions by 2050 and working back from there.

Utilities, on the other hand, are likely assuming that they'll be able to water down those targets or figure out some creative accounting to get around them. Their push for so-called renewable natural gas is just one example. PG&E, for example, cites the source of methane in its most recent R&D strategy report as part of its net-zero pathway. (Methane is a greenhouse gas that's 81 times more potent than carbon dioxide over 20 years; leaks from production and distribution infrastructure alone are a significant source of carbon emissions.)

Musk's prediction of a tripling in demand is probably an upper bound of what's possible. It shouldn't be surprising that he's aiming high — Tesla stock would surge if it came true — and an extreme take on the future is in line with his "hardcore" management style. But as electric motors and batteries get better, smaller and cheaper, they'll find their way into more things. There's a better chance than not that we'll be electrifying things we'd never thought of before. Couple that with falling renewable prices, and the world is setting itself up for a classic case of induced demand.

Like many things, the truth is probably somewhere in the middle. Musk might be overstating things, and the utilities are probably grossly understating them. But which would you rather have: utilities with stretched balance sheets or widespread blackouts?

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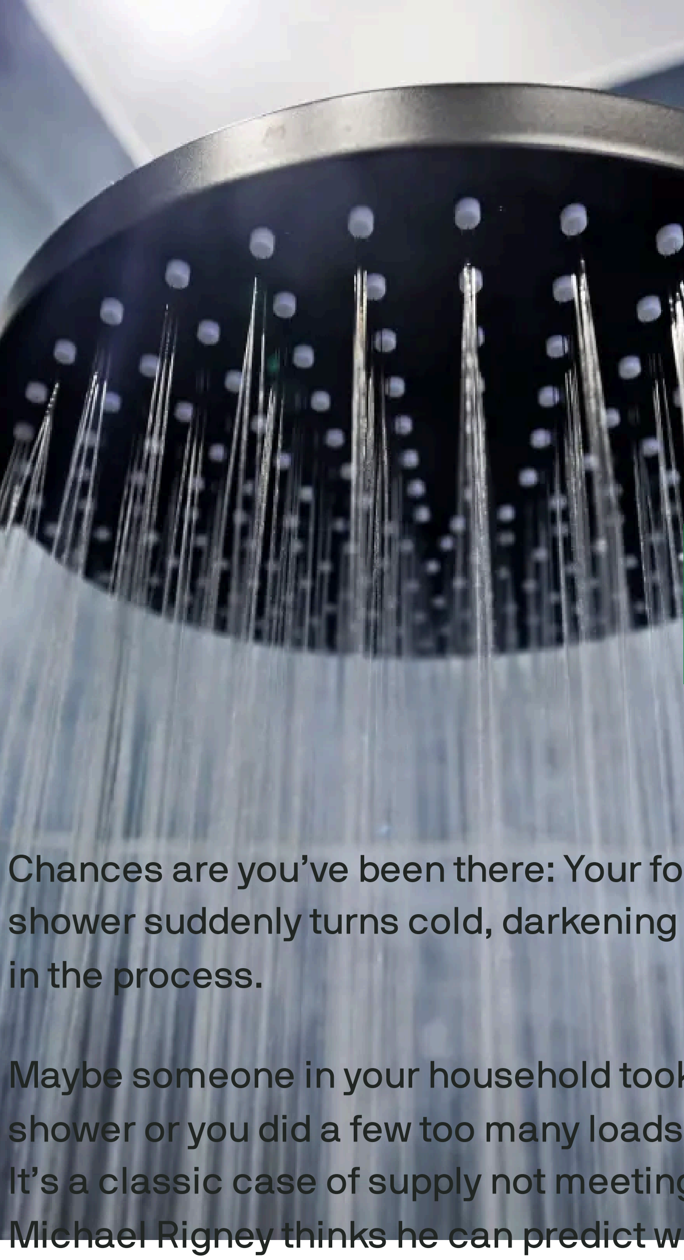

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AI-powered water heater could banish cold showers and carbon pollution

Tim De Chant 6:00 AM PDT · August 6, 2024

Chances are you've been there: Your formerly hot shower suddenly turns cold, darkening your mood in the process.

Maybe someone in your household took an extra shower or you did a few too many loads of laundry. It's a classic case of supply not meeting demand. Michael Rigney thinks he can predict when a household will need more hot water, eliminating cold showers without turning to expensive alternatives like on-demand water heaters.

Rigney's quest started a few years ago when he began researching water heating. What he saw "was really eye opening," Rigney told TechCrunch in a recent interview.

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pump water heaters were beginning to take off. Because of that shift, he said, “there was an entrepreneurial opportunity to build the best heat pump water heater.”

Water heaters tend to be fairly straightforward appliances: At their core, they have three basic parts: an insulated tank, a heating element and a thermostat. Most people set the temperature once and forget it; on rare occasions, they might set it a bit higher when overnight guests arrive. As hot water is drawn from the tank, cold water replaces it, driving down the temperature inside. When the temp gets low enough, the thermostat tells the heater to turn on.

“That’s really antiquated,” Rigney said. “We can do better in 2024.”

CALA SYSTEMS' FIRST PRODUCT WILL BE A 65-GALLON HEAT PUMP WATER HEATER.

IMAGE CREDITS:CALA SYSTEMS

Cala Systems' water heater pairs an advanced heat pump with an AI-powered control system to forecast hot water demand and heat the water in the tank accordingly. The company gathers general information like weather forecasts and time-of-use energy pricing and sends it to the water heater. That data is then analyzed on the device along with household-specific information, including water usage patterns, incoming cold water temperature and whether the home has solar panels. (Rigney said the company will never sell household data.)

By analyzing water use patterns, the tank can anticipate when there will be a surge in demand and heat the water in the most efficient way possible.

For example, if the weather is predicting a handful of sunny days before a couple days of clouds, Cala's algorithms might decide to use daytime power from a homeowner's solar panels to overheat the tank. Then, when hot water is called for, it mixes it with cold water to cool down to the appropriate temperature. That allows the water heater to make the most of excess solar production, essentially turning the tank into a battery that stores energy for a cloudy day.

In other instances, when both weather and water demand are consistent, Cala's water heater can slow down the speed of the compressor, lengthening the time it takes to heat the tank for an efficiency bump. "In water heating, when you slow down the compressor, you increase the efficiency of heat transfer by about 30%." Rigney said. "It's a pretty significant impact."

And if there are house guests coming? Cala included a boost mode that people can activate on the tank or in an app.

Today, water heaters in the U.S. are split almost evenly between natural gas and electric resistance, with oil, propane and heat pumps rounding things out. Water heating is responsible for around 20% of the typical American household's energy usage, and heat pump water heaters slash that significantly while also cutting people's dependence on natural gas.

Heat pump water heaters may only make up a few percent of the market, but they are rapidly growing in share, helped in part by incentives within the Inflation Reduction Act. And while they're more expensive to install up front, they're cheaper in the long run because they're more efficient to operate, leading to lower household carbon emissions.

Rigney said that Cala will be buying parts from various suppliers and assembling the final product in the U.S. ("This is not a product that ships well," he said with a laugh.) The company's first product, a 65-gallon model, will cost $2,850; it's available to preorder with delivery sometime early next year. That's about $800 more than competitors, though Rigney said that lower utility bills should eliminate that difference over time.

To support the rollout, Cala exclusively told TechCrunch that it has raised a $5.6 million seed round led by the Clean Energy Venture Group and the Massachusetts Clean Energy Center, with Burnt Island Ventures, CapeVista Capital and Leap Forward Ventures participating. With so few heat pump water heaters sold to date, “this is a category that is truly nascent,” Rigney said. “We think there’s an opportunity here to redefine what people expect a water heater to do.”

Topics: AI Cala Systems Clean Energy Venture Group Climate energy efficiency Exclusive heat pumps machine learning Startups water heater

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OPower Gets $50 Million To Drive Energy Efficiency Via Peer Pressure

Lora Kolodny
8:11 PM PST · November 29, 2010

The Arlington, Va. clean tech startup, OPower, closed a $50 million series C investment led by Accel Partners and Kleiner Perkins Caulfield & Byers (KPCB) and joined by the company's earlier investors New Enterprise Associates (NEA) the firms announced Monday.

OPower's software-as-a-service helps electric and gas utilities understand who their residential energy consumers are, and how they are using power. Home owners get access to OPower programs through their utilities. OPower's applications let them see if they're more or less energy-hogging than their neighbors, allow them to set personal goals to reduce their own energy consumption at home, and receive alerts if they're headed for a large bill at the end of the month among other things that are meant to inspire a behavior change.

OPower's chief executive Dan Yates jokes, "We get paid to tell people to turn out the lights and turn down the thermostat." OPower's series C deal came together in just six weeks and closed on Thanksgiving eve Yates said. Accel and KPCB contributed equal amounts.

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A partner at Accel, Peter Wagner explained his fund's decision to invest at this level in OPower:

"OPower focuses on one of the most important issues of our time— improving energy efficiency on a massive scale. [Their] model is a perfect fit with Accel's strategy for the energy sector. We focus on opportunities where information technology and the internet can create value in large-scale markets while maintaining a high degree of capital efficiency."

[*Read*: clean tech concepts that aren't as political, capital- and materials-intensive as biofuels or electric vehicles.]

Accel's portfolio is well-known for social web success stories including Facebook, Groupon and Playfish. Wagner noted:

"OPower can take advantage of the dynamics of the social web to motivate significant and lasting efficiency gains across millions of households, [while providing] deep consumer engagement that is extremely valuable to utilities."

The company will use its new-found capital Yates said, "To radically expand our R&D effort, bring 50 more people on board, and above all to build more product." The CEO would not elaborate on what specific features and applications the company sought to release in 2011. He said most of the new product would be designed for energy consumers, rather than utilities. (OPower also provides utilities with a system to help customer service reps manage relationships and handle inquiries.)

Currently, OPower's 45 clients are all U.S.-based utilities. OPower has had lots of inbound inquiries, Yates said, including from foreign utilities beyond North America, but hasn't decided where to set up overseas operations.

The approach of using peer-pressure to drive more environmentally responsible behavior at home could work well in nations like South Korea, where the cost of trash collection goes up for an entire community when just a few people within it— including embarrassing Americans— fail to keep recycling and refuse separated.

Topics: Accel Climate Dan Yates energy efficiency GreenTech kpcb nea OPOWER Peter Wagner series C

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